MemoryOS: Software Requirement Specification

Date: May 1, 2026 | **Status:** Final Executive Spec

Strategic Lead: Analyst / PM Office | **Technical Lead:** CTO Office

1. Executive Summary

MemoryOS is a **Private Personal Memory Operating System**. It shifts the paradigm from passive note-taking to active synthesis. By leveraging local-first AI and vector databases on the edge, it creates a searchable, relational, and 100% private digital twin of the user's experiences.

2. Strategic Pillars (PM Review)

2.1 Core Problem: The Decay of Context

Information is easy to capture but context decays rapidly. MemoryOS preserves context by linking entries semantically rather than just chronologically.

2.2 The Privacy Moat

Data never leaves the device. This "Zero-Knowledge" approach allows users to record sensitive thoughts, medical logs, and professional secrets without fear of data mining or breaches.

3. Technical Architecture (CTO Spec)

Component	Technology Stack	Functional Purpose
Core Engine	React Native + JSI	

Component	Technology Stack	Functional Purpose
		Native-speed JS/C++ bridging for UI and AI.
Semantic Store	LanceDB (Embedded)	Local vector database for HNSW-based RAG.
AI Inference	Transformers.js / ONNX	On-device embedding generation (Zero-cloud).
Security	AES-256-GCM + Keychain	Hardware-backed encryption at rest.

4. High-Level Requirements

- **Natural Language Recall:** Users must be able to query the app using conversational language (e.g., "Why did I decide against that new job in February?").
- **Proactive Synthesis:** The system should analyze weekly trends and present "Insight Cards" without user prompting.
- **Omni-Ingestion:** Support for high-speed voice transcriptions using local Whisper models.

5. Roadmap

- **Phase 1:** Local-first database infra and basic vector search (Month 1).
- **Phase 2:** Integration of small language models (SLM) for on-device reasoning (Month 2).
- **Phase 3:** Proactive notification engine and encrypted cross-device sync (Month 3).